A REPORT
ON
Analysis of Air-Quality and Forecasting

BY

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(Nov, 2024)

1. INTRODUCTION TO AQI AND ITS CALCULATION

National Air Quality Index

1. The Air Quality Index is a tool for effectively communicating air quality status to people in terms that are easy to understand. It transforms complex air quality data of various pollutants into a single number (index value), vocabulary and colour.

2. There are six AQI categories: Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe. Each category is decided based on ambient concentration values of air pollutants and their likely health impacts (known as health breakpoints). AQ sub-index and health breakpoints are evolved for eight pollutants (PM10, PM2.5, NO2, SO2, CO, O3, NH3, and Pb) for which short-term (up to 24-hours) National Ambient Air Quality Standards are prescribed.

3. Based on the measured ambient concentrations of a pollutant, a sub-index is calculated, which is a linear function of concentration (e.g. the sub-index for PM2.5 will be 51 at a concentration of 31 µg/m3, 100 at a concentration of 60 µg/m3, and 75 at a concentration of 45 µg/m3). The worst sub-index determines the overall AQI. AQI categories and health breakpoints for the eight pollutants are as follows:

AQI Category	AQI	Concentration Range*							
		PM_{10}	$PM_{2.5}$	NO_2	O_3	CO	SO_2	NH_3	Pb
Good	0-50	0-50	0-30	0–40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory	51 - 100	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.5-1.0
Moderately Polluted	101-200	101-250	61-90	81-180	101-168	2.1-10	81-380	401-800	1.1-2.0
Poor	201-300	251-350	91-120	181-280	169-208	10-17	381-800	801-1200	2.1-3.0
Very Poor	301–400	351-430	121-250	281-400	209-748*	17-34	801-1600	1200-1800	3.1-3.5
Severe	401-500	430+	250+	400+	748+*	34+	1600+	1800+	3.5+

* CO in mg/m^3 and other pollutants in $\mu g/m^3$; 24-hourly average values for PM_{10}, $PM_{2.5}$, NO_2, SO_2, NH_3, and Pb, and 8-hourly values for CO and O_3.

How is AQI Calculated?

1. The Sub-indices for individual pollutants at a monitoring location are calculated using their 24-hour average concentration value (8-hourly in the case of CO and O3) and health breakpoint concentration range. The worst sub-index is the AQI for that location.

2. All eight pollutants may not be monitored at all the locations. Overall, AQI is calculated only if data are available for a minimum of three pollutants, out of which one should necessarily be either PM2.5 or PM10. Otherwise, data are considered insufficient for calculating AQI. Similarly, a minimum of 16 hours of data is deemed necessary to calculate subindex.

3. The sub-indices for monitored pollutants are calculated and disseminated, even if data are inadequate for determining AQI. The individual pollutant-wise sub-index will provide the air quality status of that pollutant.

4. The web-based system is designed to provide AQI in real-time. It is an automated system that captures data from continuous monitoring stations without human intervention and displays AQI based on running average values (e.g. AQI at 6 am on a day will incorporate data from 6 am on the previous day to the current day).

5. For manual monitoring stations, an AQI calculator is developed wherein data can be fed manually to get an AQI value.

AQI Calculation Using Spreadsheet XL

AQI for a particular day and at a desired location can be calculated using MS Excel (As shown in the figure), wherein a user-friendly evaluation of AQI has been developed. The user needs to input at least three values of pollutant concentration (including at least one of PM10 or PM2.5) in the blue cells, and the sub-indices are calculated, thus displaying the final AQI along with the colour signifying the AQI category. The health impacts corresponding to the AQI category are detailed in the legend at the bottom of the sheet. This XL program can be obtained from CPCB.

Overall, AQI is calculated only if data are available for a minimum of three pollutants, out of which one should necessarily be either PM2.5 or PM10. Otherwise, data are considered insufficient for calculating AQI. Similarly, a minimum of 16 hours of data is necessary for calculating the subindex.

2. EXPLORATORY DATA ANALYSIS (EDA)

Data Collection and Preprocessing

Hourly AQI data for Jaipur and Kolkata (2018–2023) was sourced from the [Central Pollution Control Board (CPCB) AQI Repository](#).

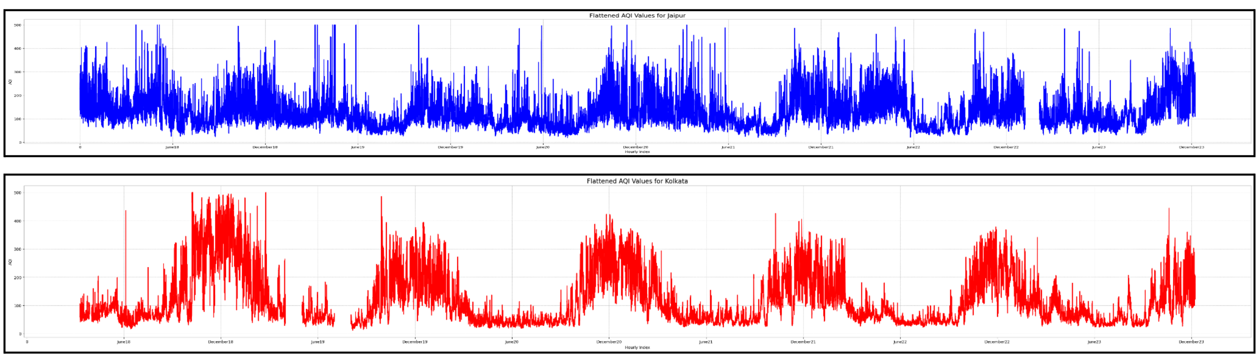
The dataset is structured with:

- **Rows**: Representing consecutive days.
- **Columns**: 24 columns for hourly AQI measurements (one for each hour) and an additional column for the date.

	Date	00:00:00	01:00:00	02:00:00	03:00:00	04:00:00	05:00:00	06:00:00	07:00:00	08:00:00	...	14:00:00	15:00:00	16:00:00	17:00:00	18:00:00	19:00:00	20:00:00	21:00:00	22:00:00	23:00:00
0	1	226.0	228.0	234.0	253.0	218.0	220.0	137.0	148.0	161.0	...	150.0	134.0	111.0	110.0	112.0	120.0	157.0	226.0	238.0	251.0
1	2	257.0	236.0	229.0	192.0	212.0	227.0	229.0	243.0	257.0	...	163.0	140.0	130.0	126.0	120.0	148.0	192.0	226.0	239.0	269.0
2	3	273.0	266.0	263.0	265.0	260.0	250.0	230.0	227.0	255.0	...	201.0	164.0	141.0	127.0	134.0	203.0	304.0	349.0	404.0	375.0
3	4	313.0	276.0	253.0	224.0	182.0	160.0	140.0	158.0	200.0	...	119.0	137.0	149.0	105.0	136.0	142.0	191.0	235.0	236.0	211.0
4	5	197.0	173.0	165.0	151.0	140.0	145.0	153.0	160.0	200.0	...	144.0	158.0	134.0	126.0	146.0	199.0	238.0	272.0	299.0	300.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2186	27	226.0	207.0	189.0	180.0	159.0	170.0	206.0	191.0	253.0	...	123.0	125.0	114.0	107.0	123.0	169.0	340.0	325.0	272.0	249.0
2187	28	259.0	257.0	229.0	204.0	230.0	243.0	292.0	278.0	279.0	...	153.0	145.0	141.0	128.0	162.0	224.0	309.0	317.0	294.0	308.0
2188	29	325.0	327.0	323.0	327.0	340.0	323.0	301.0	302.0	285.0	...	198.0	179.0	124.0	157.0	131.0	141.0	206.0	182.0	192.0	200.0
2189	30	203.0	213.0	215.0	220.0	188.0	170.0	154.0	158.0	168.0	...	194.0	162.0	162.0	177.0	186.0	205.0	262.0	287.0	276.0	222.0
2190	31	242.0	234.0	229.0	240.0	236.0	230.0	233.0	222.0	226.0	...	234.0	194.0	109.0	115.0	174.0	212.0	270.0	286.0	297.0	294.0

2191 rows × 25 columns

Dataframe Format



Hourly AQI data for Jaipur and Kolkata (with missing data)

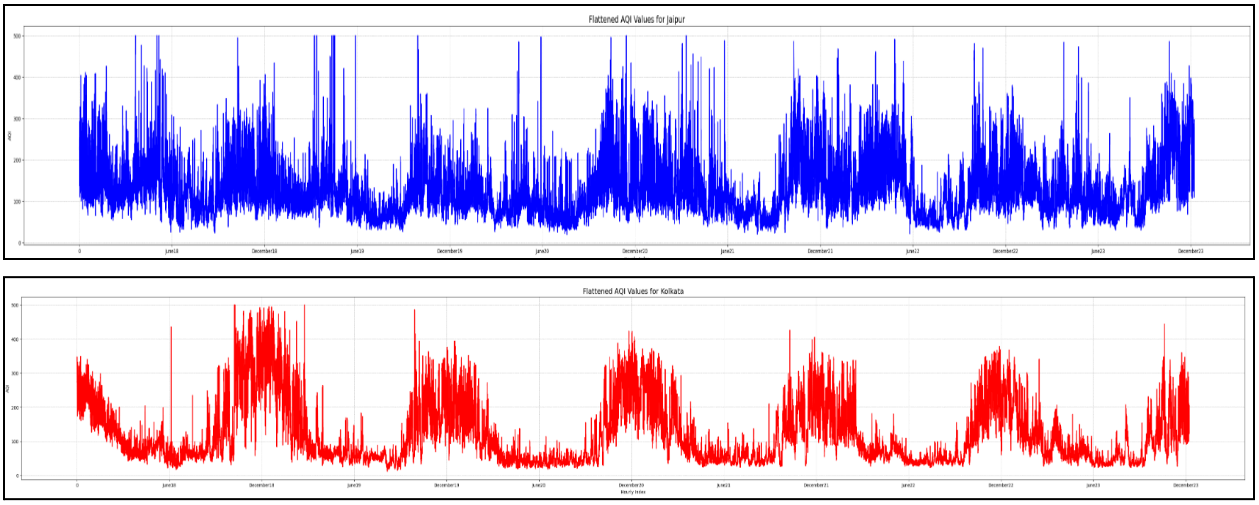
Handling Missing Data

The dataset revealed missing AQI values for:

- **Jaipur**: 1 month.
- **Kolkata**: 2 months.

To fill these gaps, the missing values were replaced with the average AQI for the exact dates and months from other years. This approach ensured the data remained consistent and reliable.

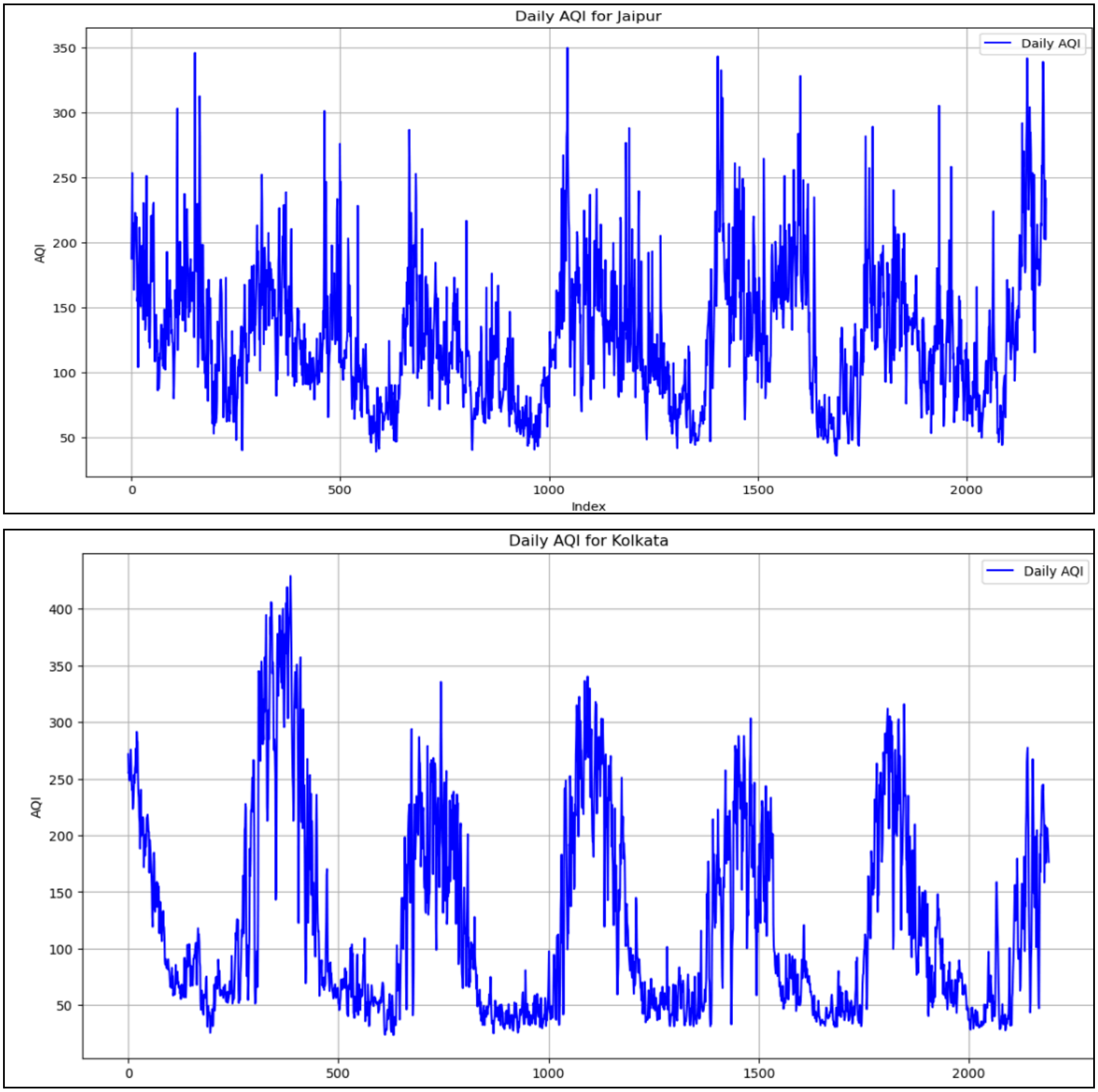
Data Visualization



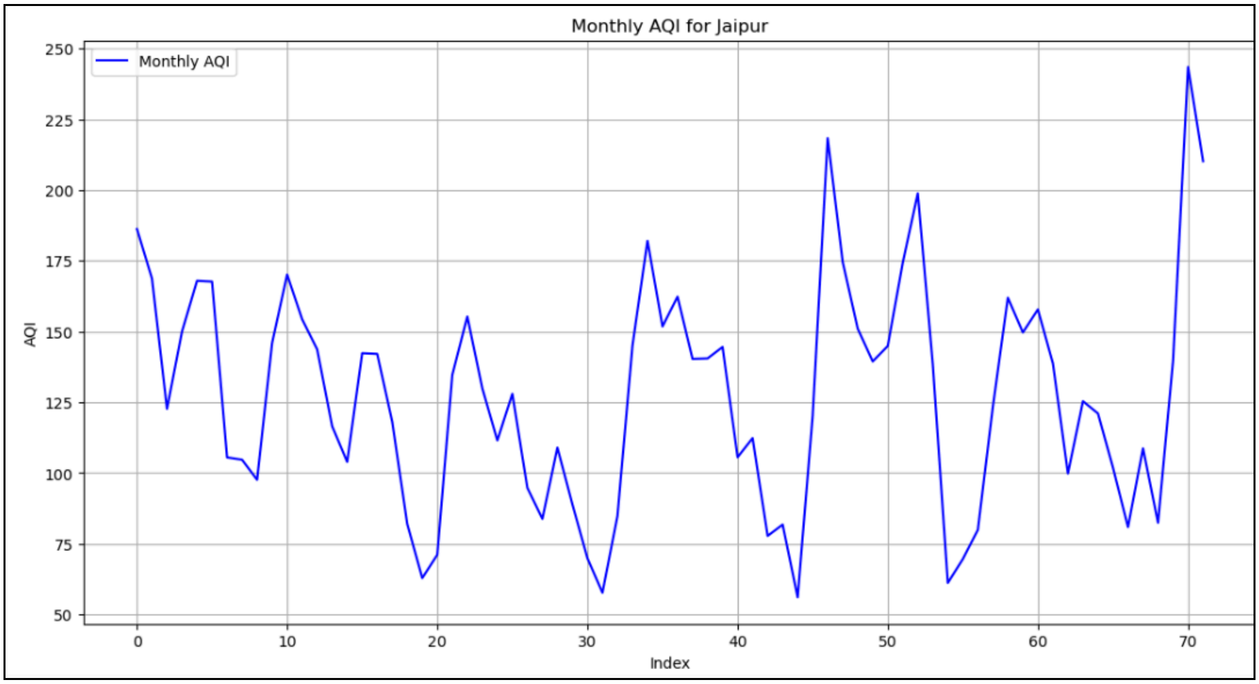
Hourly AQI data of Jaipur and Kolkata

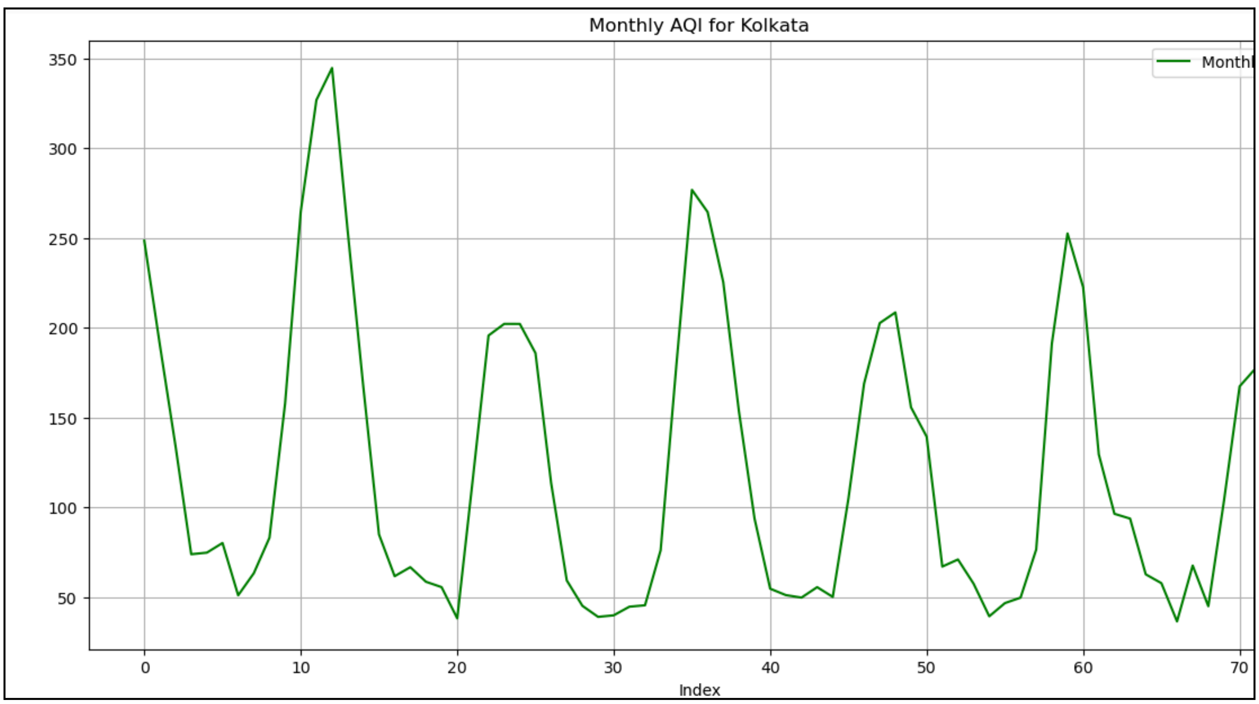
To get a clear picture of the AQI trends in both cities, we looked at the data on a daily, monthly, and yearly basis.

- For **daily AQI**, we averaged the hourly values for each day.
- **Monthly AQI** was calculated by taking the average daily AQI values in a month.
- Finally, **yearly AQI** was found by averaging the monthly AQI values for each year.



Daily AQI data for Jaipur and Kolkata





Monthly AQI data for Jaipur and Kolkata

Inference for Daily and Monthly AQI Data

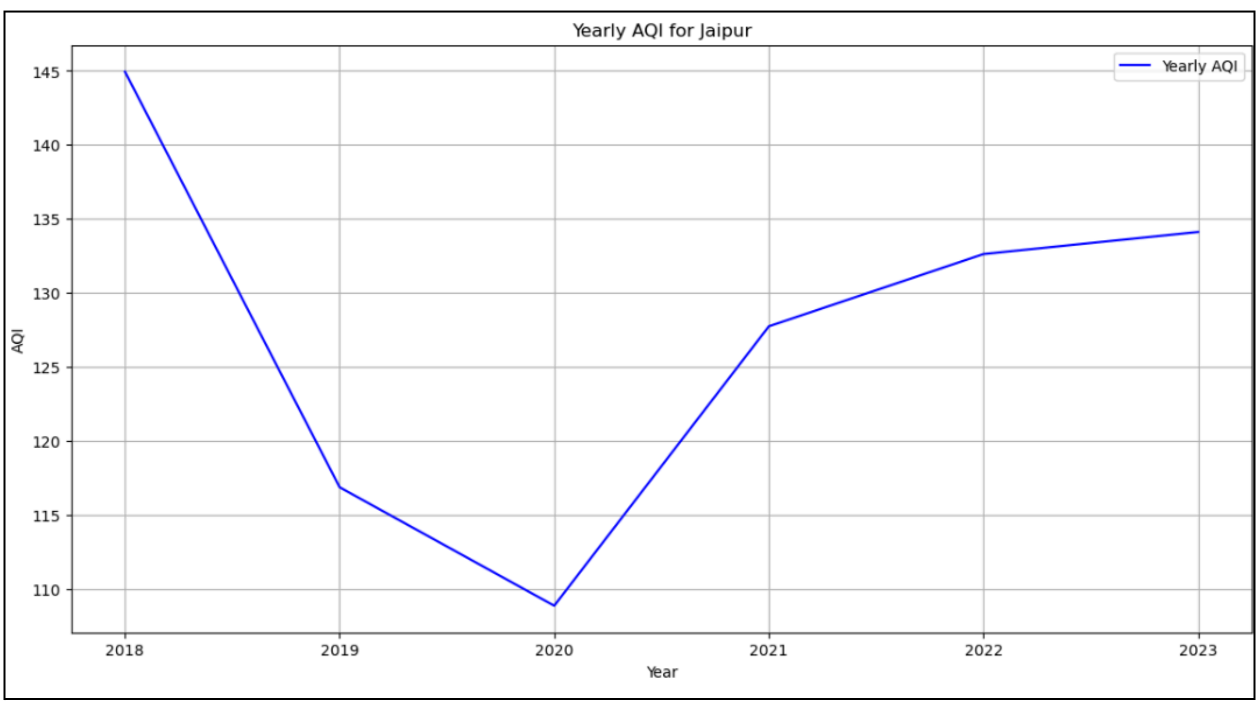
Jaipur: Stability with Seasonal Variations

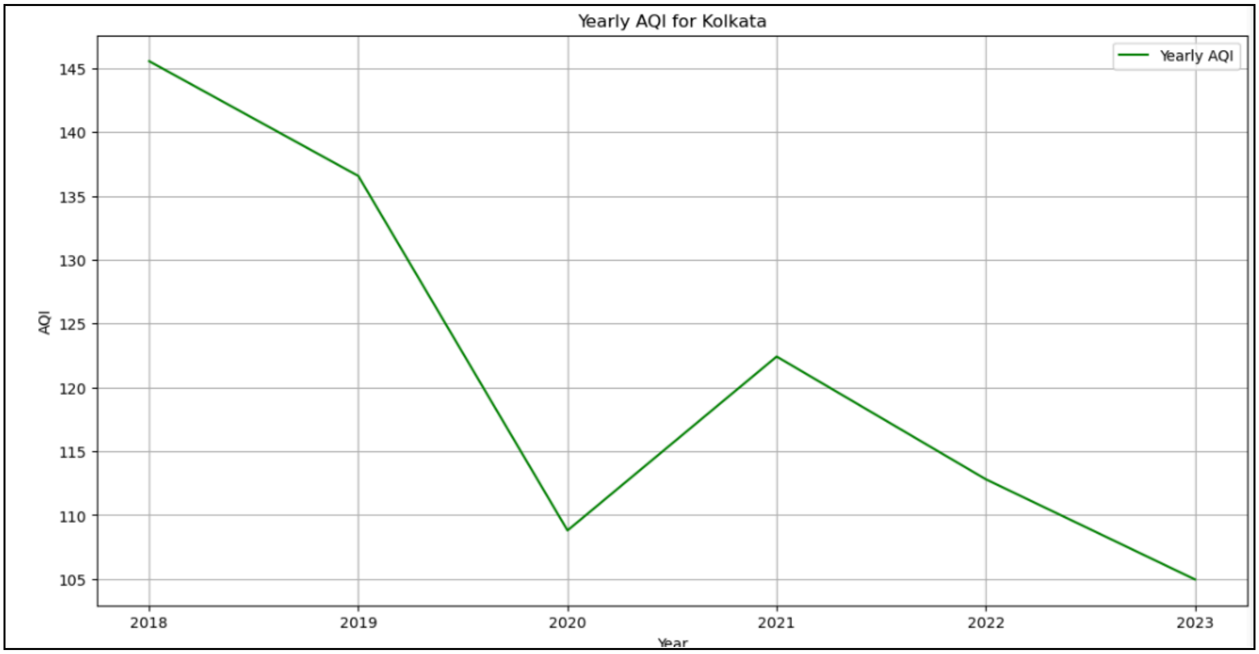
- Overall Stability: Jaipur's AQI shows fewer fluctuations over time, maintaining a relatively stable pattern compared to Kolkata.

- Winter and Summer Peaks: Air pollution tends to rise in the winter and summer, while the monsoon season improves air quality.
- Factors Influencing Air Quality in Jaipur:
    - Weather Conditions: Air quality and weather parameters like temperature, humidity, and wind speed are strongly linked.
    - Nitrogen Dioxide (NO_2): NO_2 concentrations peak between February and March, contributing to seasonal pollution.
    - Cold Air in Winter: Dense, slow-moving cold air traps pollutants closer to the surface, leading to prolonged pollution exposure.
    - Hot Summers: High temperatures promote the formation of ground-level ozone, adding to pollution levels.

Kolkata: Clear Seasonal Trends

- Seasonal Spikes: Kolkata experiences noticeable AQI increases around December, reflecting well-defined seasonal patterns.
- Key Contributing Factors:
    - Fuel Combustion: Increased fuel burning for heating and transportation during winter contributes to higher pollution levels.
    - Cold Air: Similar to Jaipur, cold, dense air traps pollutants, exacerbating winter pollution.
    - Firecrackers: Post-Diwali festivities, including firecrackers, significantly contribute to pollution.





Yearly AQI data for Jaipur Kolkata

Inference for Yearly AQI Data

2020 Dip in AQI: Both cities experienced a noticeable dip in yearly AQI during 2020, which can be attributed to the reduced human activity and emissions during the COVID-19 pandemic lockdowns.

Post-2020 Trends:

- Jaipur: After 2021, AQI levels began to rise again, reflecting a gradual return to pre-pandemic activity.
- Kolkata: Unlike Jaipur, Kolkata's AQI showed a further decrease after 2021, indicating potential improvements in air quality management or the continued effects of reduced emissions.

3. TEST FOR NORMAL DISTRIBUTION

Using the hourly AQI data for Jaipur and Kolkata, the following statistics were calculated to check whether the distribution over a day is normally distributed. The number of rows indicates how often the data was identified as normally distributed according to the three tests described below.

- **Shapiro-Wilk Normality:** Evaluates whether a dataset deviates significantly from a normal distribution and is sensitive to both skewness and kurtosis.
- **Kolmogorov-Smirnov Normality:** Compares the dataset to a specified distribution and assesses the maximum deviation between their cumulative distribution functions.
- **Anderson-Darling Normality:** Extends the Kolmogorov-Smirnov test, giving more weight to the distribution's tails making it stricter for detecting deviations from normality.

Next, the AQI data was fitted to various statistical distributions to identify which models align most closely with its behaviour. The counts indicate how often each distribution was recognised as the best fit. The distributions used were **Exponential, Weibull, Normal, Gamma** and **Lognormal.**

1. **JAIPUR**

```
Total valid rows analyzed: 2191
Normal by Shapiro-Wilk: 1098
Normal by Kolmogorov-Smirnov: 2048
Normal by Anderson-Darling: 1127
Best-Fit Distribution Counts:
expon: 322
weibull_min: 840
norm: 205
gamma: 243
lognorm: 581
```

Test Results for Jaipur's Daily AQI

Inference for Normality in Jaipur's Daily AQI

- **Shapiro-Wilk Normality:** About 1,098 rows were classified as normal. This implies about **50%** of the data appears to conform to a normal distribution, indicating a partial alignment with Gaussian characteristics.

- **Kolmogorov-Smirnov Normality:** About 2,048 rows were classified as normal.This implies a much higher percentage (**~93%**) of data is considered normal by this test, which is less strict than others and more suitable for larger datasets.
- **Anderson-Darling Normality:** About 1,127 rows were classified as normal. According to this more stringent test, approximately **51%** of the data fits a normal distribution.

These results suggest that while the AQI data exhibits some normality, it is not strictly Gaussian. This partial normality may reflect the influence of underlying seasonal and event-driven spikes or dips in AQI levels.

Inference for Other Distributions in Jaipur's Daily AQI

- **Exponential** (322 rows): A small proportion (~15%) of the data exhibits exponentially distributed characteristics, indicating instances of rapidly decaying AQI levels.
- **Weibull** (840 rows): The most common distribution (~38%) suggests that AQI levels in Jaipur often follow a Weibull distribution, which can model events where values cluster around a lower bound with occasional outliers.
- **Normal** (205 rows): Only ~9% of data fits a normal distribution. This low count corroborates the earlier findings that AQI levels deviate significantly from Gaussian behaviour.
- **Gamma** (243 rows): About ~11% of the data follows a Gamma distribution, often used to model skewed and positive-only data such as AQI.
- **Lognormal** (581 rows): A significant portion (~26%) follows a log-normal distribution, suggesting that AQI levels are multiplicative, with occasional high spikes that deviate substantially from the mean.

2. Kolkata

```
Summary of AQI Distribution Analysis:
Total valid rows analyzed: 2191
Normal by Shapiro-Wilk: 1042
Normal by Kolmogorov-Smirnov: 2082
Normal by Anderson-Darling: 1038
Best-Fit Distribution Counts:
weibull_min: 1059
norm: 150
expon: 349
lognorm: 413
gamma: 220
```

Test Results for Kolkata's Daily AQI

Inference for Normality in Kolkata's Daily AQI

- **Shapiro-Wilk Normality:** This indicates that a significant portion of the data (1042 out of 2191), or 47%, is close to a normal distribution, but not the majority, as the test is quite sensitive to deviations.
- **Kolmogorov-Smirnov Test:** Suggests that the AQI data largely follows a normal distribution according to this test, about 95%, as 2082 out of 2191 values pass.
- **Anderson-Darling Test:** Similar to the Shapiro-Wilk test, which is about 47%, this indicates that a substantial portion of the data appears normal, although it's stricter on tail behaviour.

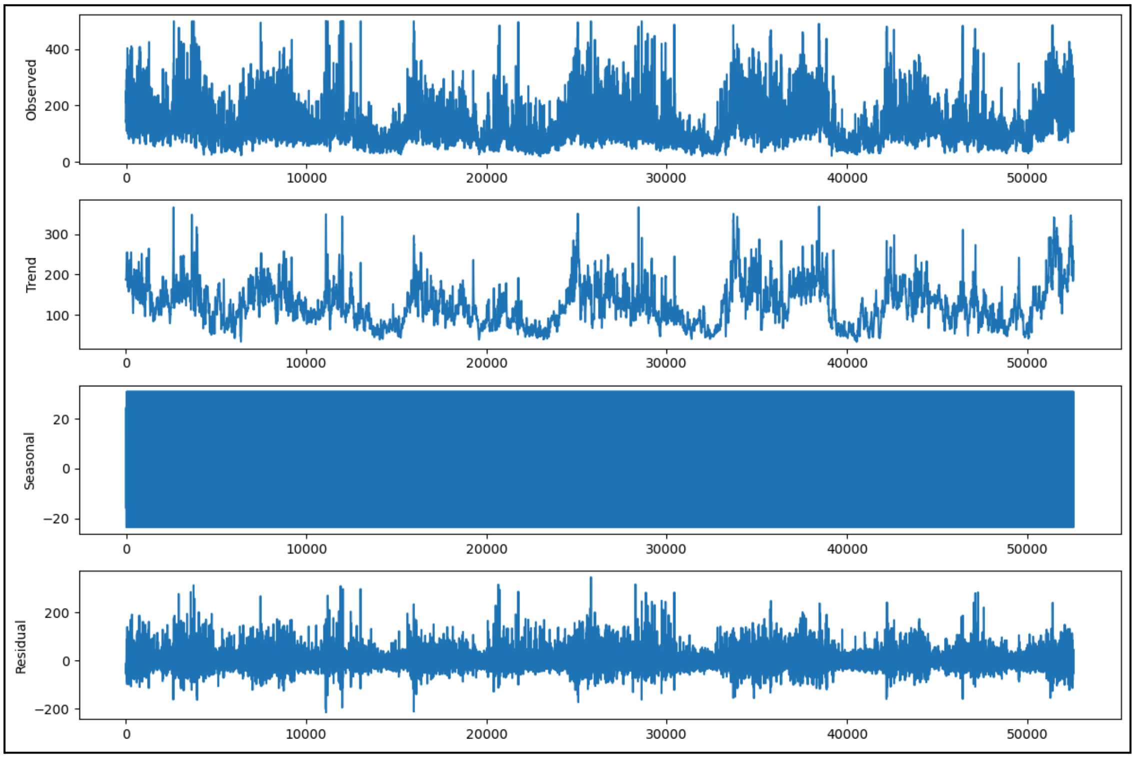
The results are similar to that obtained while analysing Jaipur's data. These results suggest that while the AQI data exhibits some normality, it is not strictly Gaussian. This partial normality may reflect the influence of underlying seasonal and event-driven spikes or dips in AQI levels.

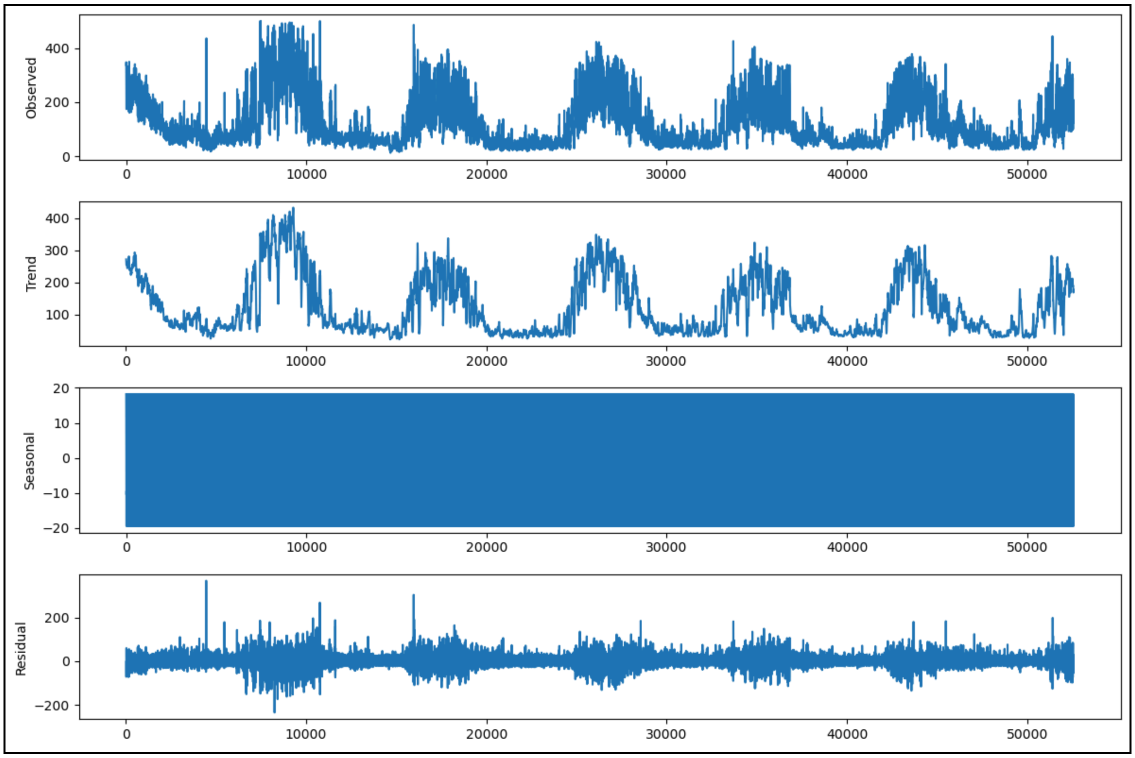
Inference for Other Distribution in Kolkata's Daily AQI

- **Exponential** (349 rows): This distribution fits a moderate portion (~16%) of the data, indicating possible random events or processes in the AQI readings.
- **Weibull_Min** (1059): (~48%) The highest number of data points best fits the Weibull distribution, which typically suggests a skewed distribution with varying spread, potentially indicative of skewed environmental factors like weather conditions or seasonal events.
- **Normal Distribution** (150)**:** A smaller portion of data fits a normal distribution, reinforcing that the data doesn't fully conform to normality, though some periods might. (~6&)
- **Gamma Distribution** (220): The gamma distribution suggests that some AQI values are positively skewed, and the distribution might be useful in modelling pollutant accumulation or extreme events. (~10%)
- **Log-Normal Distribution** (413)**:** The log-normal distribution fits a significant portion, often indicating that the AQI levels follow a multiplicative process. (~19%)

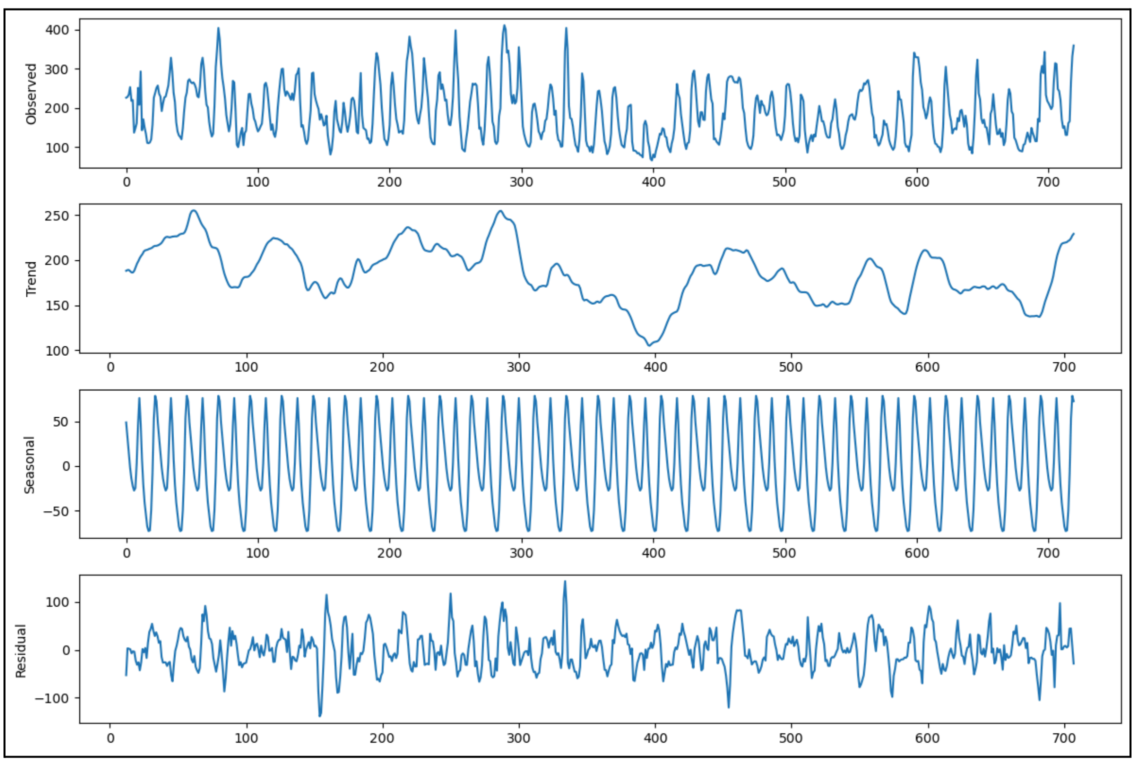
4. TREND AND SEASONALITY IN THE AQI DATA:

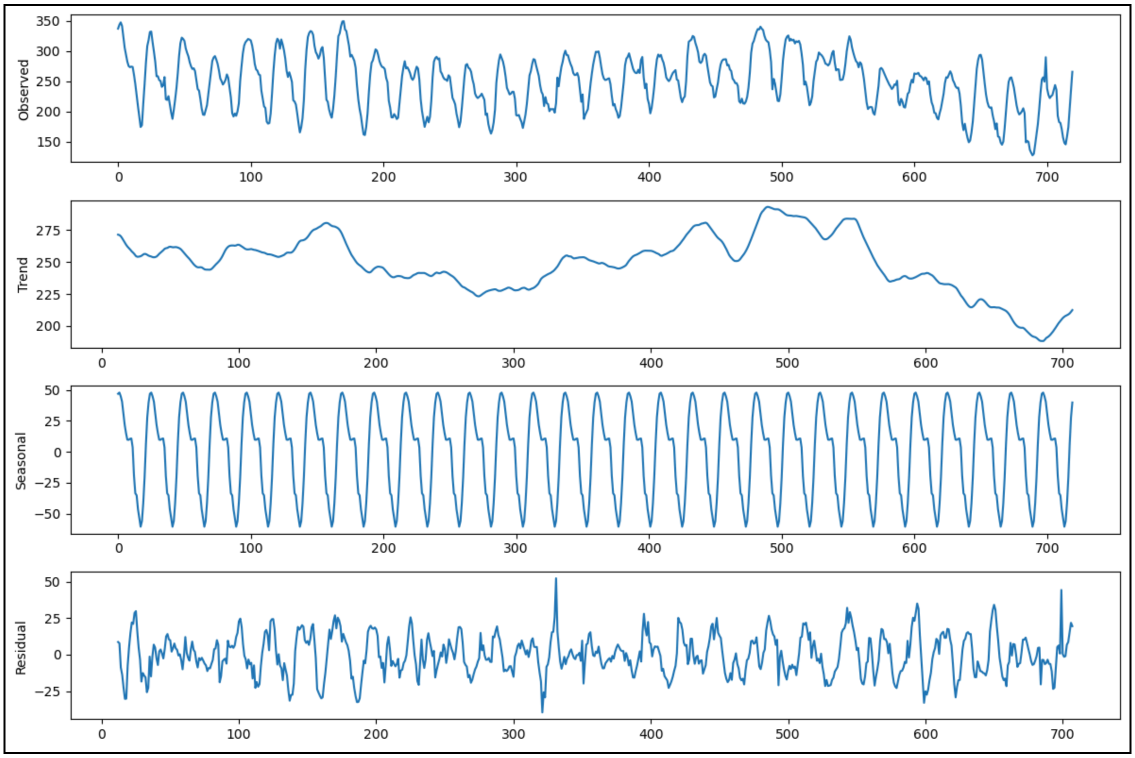
Daily Plots





Decomposed Time Series AQI Data for Jaipur and Kolkata (Period = 24)

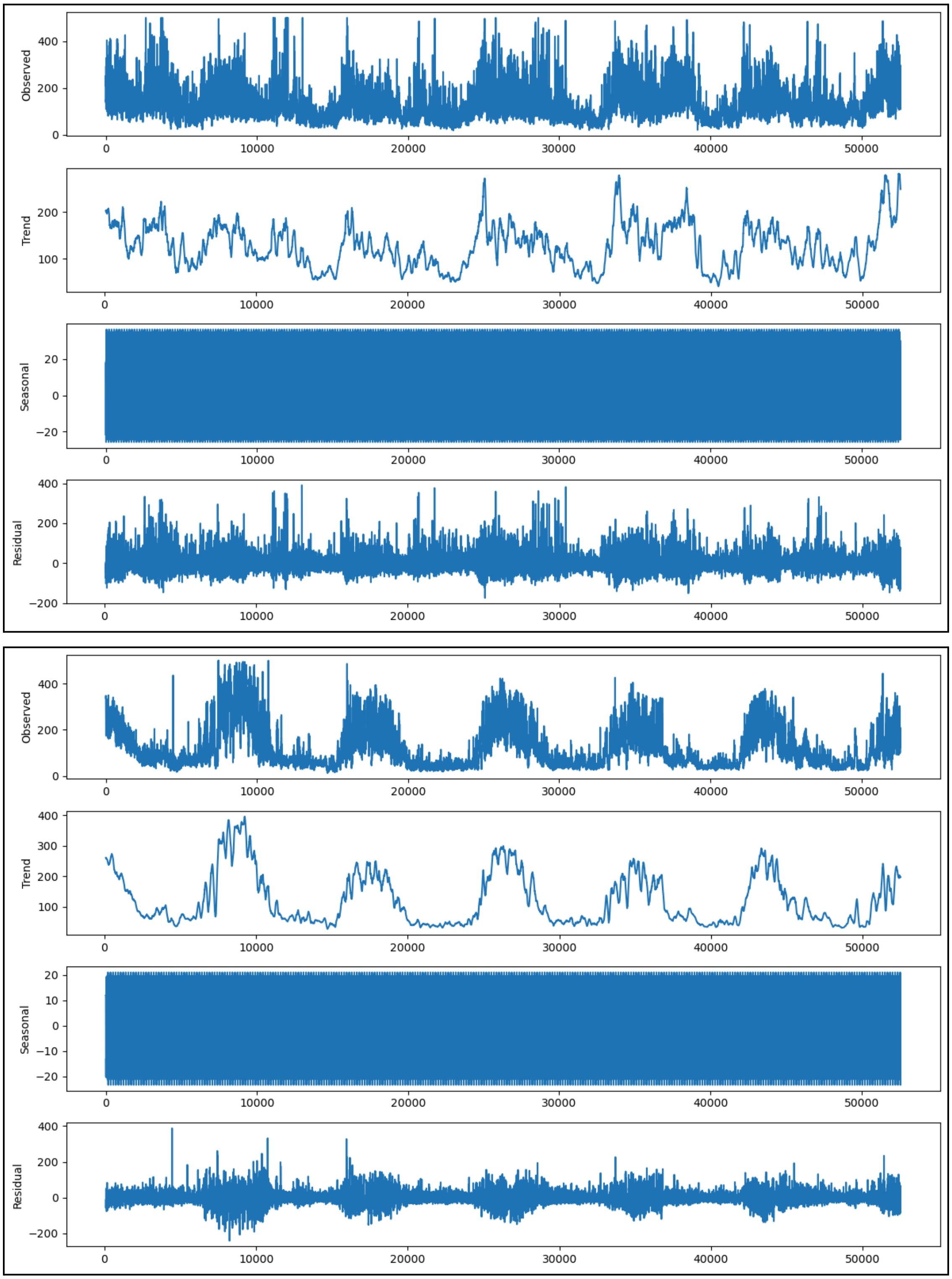




***Scaled Up** Decomposed Time Series AQI Data for Jaipur and Kolkata (Period = 24)*

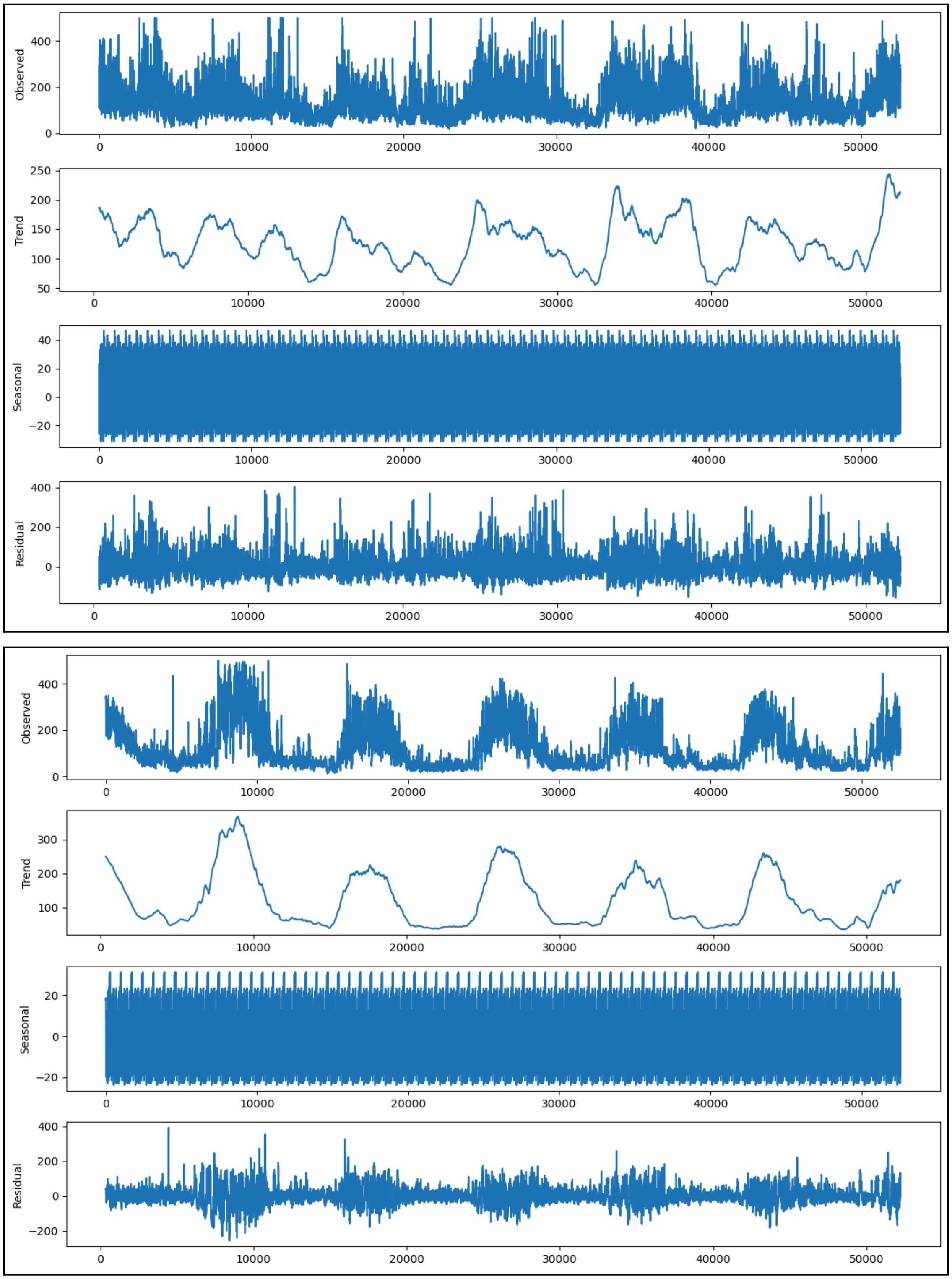
These two plots are identical to the previous ones but have been scaled up for better clarity. They now display the seasonal data, providing a clearer view of the trends and variations in AQI.

Weekly Plots



*Decomposed Time Series AQI Data for Jaipur and Kolkata (Period = 24*7)*

Monthly Plots



*Decomposed Time Series AQI Data for Jaipur and Kolkata (Period = 24*30)*

Choice of Period:

- Daily (Period = 24), Weekly (Period = 24 × 7), and Monthly (Period = 24 × 30) analyses are less suitable due to:
    1. Lack of Clear Trends or Seasonality:
        - These periods are too short to capture long-term patterns or the full extent of seasonal cycles.
        - The high noise in the data overshadows the underlying trends and seasonal components, leading to ambiguous results.
    2. Short-Term Fluctuations:
        - These periods primarily reflect short-term variability in AQI due to daily activities or weekly cycles, which do not provide meaningful insights into long-term air quality changes.

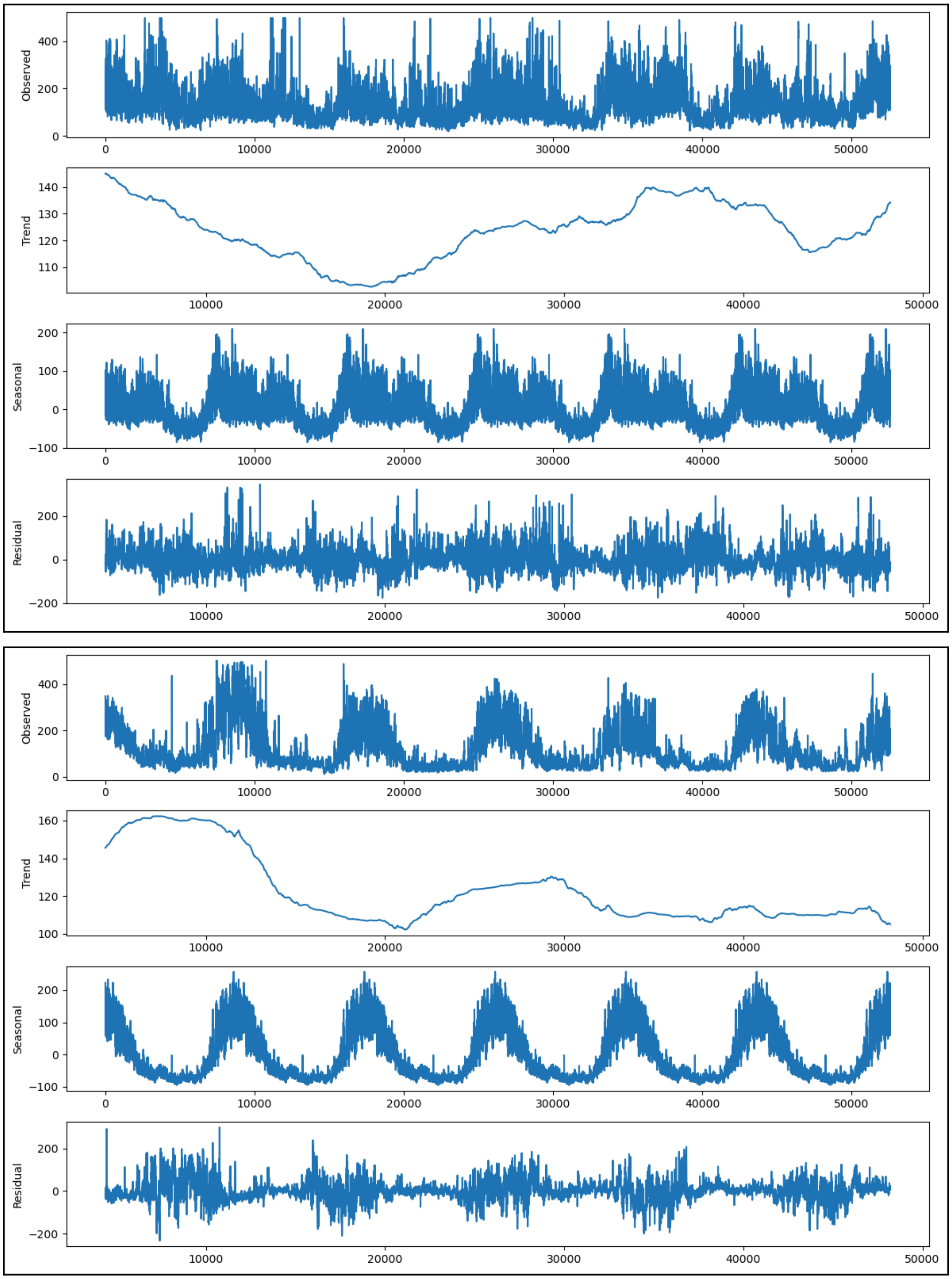
Trends Analysis

- **Jaipur:**
    - Despite the noise, Jaipur shows an increase in the trend component, suggesting a potential worsening of air quality over time.
    - This indicates that urbanization, industrial activity, or other local factors may be driving pollution levels higher.
- **Kolkata:**
    - The trend for Kolkata shows a dip, which may reflect temporary improvements due to factors like the COVID-19 lockdowns or other interventions.
    - However, the absence of a clear, sustained pattern highlights the limitations of using shorter periods for meaningful trend analysis.

Seasonality Analysis

- While some seasonality is visible in these periods, it is often incomplete or inconsistent, making it difficult to identify robust seasonal patterns or periodicity.

Yearly Plot



*Decomposed Time Series AQI Data for Jaipur and Kolkata (Period = 24*365)*

Trend Analysis

1. **Initial Decline in AQI Levels (COVID-19 Impact)**:
   - In both Jaipur and Kolkata, the trend shows a long-term decline in AQI levels during 2020, attributed to reduced human activity during the COVID-19 lockdowns.
   - This reflects the direct impact of limited vehicular movement, industrial operations, and construction activity, highlighting the significant role of human activities in driving pollution levels.
2. **Recent Upward Trajectory (Post-COVID)**:
   - Following the COVID-19-related dip, the trend reveals a gradual rise in AQI levels, suggesting a return to pre-pandemic activity levels and possibly even higher pollution levels due to increased urbanization, industrial activity, or inadequate environmental regulations.
   - This upward trend points to a potential degradation in air quality, calling for immediate policy action to curb emissions and implement sustainable practices.

Seasonality Analysis

1. **Consistent and Regular Cycles**:
   - The seasonal component exhibits highly regular and repeating yearly patterns, highlighting the predictable influence of seasonal factors such as:
     - Winter months with higher AQI levels due to stagnant air and temperature inversions trapping pollutants.
     - Monsoon season, where rainfall improves air quality by washing out particulate matter.
   - These patterns underscore the role of natural and recurring events in shaping AQI fluctuations.
2. **Human and Environmental Factors**:
   - The seasonal cycles also align with human activities, such as increased emissions during festivals, crop residue burning, or industrial activities during specific times of the year

Residual Analysis

1. **Reduced Noise in Yearly Data:**
   - When analyzing the time series with a yearly periodicity (24×365), the residual component is significantly reduced compared to weekly or monthly periods.
   - This reduction in noise suggests that a yearly period better captures long-term trends and seasonal patterns, minimizing random fluctuations and outliers.

5. STATIONARITY:

Stationarity in time series refers to a property where the statistical characteristics of the data (mean, variance, and autocorrelation) do not change over time. In simpler terms, the data behaves consistently and predictably without showing trends or seasonality.

- A stationary time series makes it easier to model and forecast because it doesn't vary in unexpected ways.
- Non-stationary data (with trends or seasonality) often requires transformation to become stationary before analysis.

To determine whether the hourly AQI data for Jaipur and Kolkata is stationary, two statistical tests—ADF (Augmented Dickey-Fuller) and KPSS (Kwiatkowski-Phillips-Schmidt-Shin)—are applied both before and after transforming the data.

1. **Augmented Dickey-Fuller (ADF) Test**:
   - **Purpose**: Checks for the presence of a **unit root**, which indicates non-stationarity.
   - **Null Hypothesis (H_0)**: The data is **non-stationary** (i.e., it has a unit root).
   - **Alternate Hypothesis (H_1)**: The data is **stationary**.
   - **Outcome**: If the p-value is low (typically < 0.05), we reject the null hypothesis and conclude the series is stationary.

2. **KPSS (Kwiatkowski-Phillips-Schmidt-Shin) Test**:
   - **Purpose**: Checks for **stationarity around a deterministic trend**.
   - **Null Hypothesis (H_0)**: The data is **stationary**.
   - **Alternate Hypothesis (H_1)**: The data is **non-stationary**.
   - **Outcome**: If the p-value is low (typically < 0.05), we reject the null hypothesis and conclude the series is non-stationary.

Before Differencing (Raw Data)

1. **Jaipur:**

```
ADF Statistic: -11.718411059167801
ADF p-value: 1.4326859700530826e-21
Reject the null hypothesis: The series is stationary.

KPSS Statistic: 0.7193165175193232
KPSS p-value: 0.01178940749824334
Reject the null hypothesis: The series is non-stationary.
```

ADF Test:

- The ADF test indicates that the data is stationary (p-value < 0.05).
- This implies the absence of a unit root and no significant trend in the raw data.

KPSS Test:

- The KPSS test indicates that the data is non-stationary (p-value < 0.05).
- This suggests that the data contains either a deterministic trend or seasonality that prevents it from being stationary.

2. **Kolkata:**

```
ADF Statistic: -7.485070214264995
ADF p-value: 4.66231754959477e-11
Reject the null hypothesis: The series is stationary.

KPSS Statistic: 1.278749771414931
KPSS p-value: 0.01
Reject the null hypothesis: The series is non-stationary.
```

ADF Test:

Similar to Jaipur, the ADF test suggests that the data is stationary (p-value < 0.05).

KPSS Test:

The KPSS test, however, concludes that the data is non-stationary (p-value < 0.05), indicating the presence of a trend or seasonal components.

After Differencing (Transformed Data)

Differencing is performed to remove trends or seasonality by subtracting each data point from its predecessor. This transformation ensures the statistical properties (mean, variance, etc.) remain constant over time.

1. **Jaipur:**

```
ADF Statistic: -37.917991871253285
ADF p-value: 0.0
Reject the null hypothesis: The series is stationary.
KPSS Statistic: 0.04411880761438895
KPSS p-value: 0.1
Fail to reject the null hypothesis: The series is stationary.
```

After applying differencing, both the ADF and KPSS tests agree that the data is now **stationary:**

- The ADF test rejects the null hypothesis of non-stationarity.
- The KPSS test fails to reject the null hypothesis of stationarity.

2. **Kolkata:**

```
ADF Statistic: -44.23689765654193
ADF p-value: 0.0
Reject the null hypothesis: The series is stationary.
KPSS Statistic: 0.04530191192332634
KPSS p-value: 0.1
Fail to reject the null hypothesis: The series is stationary.
```

Similarly, after differencing, both tests conclude that the series is **stationary**:

- The ADF test confirms stationarity by rejecting its null hypothesis.
- The KPSS test supports stationarity by failing to reject its null hypothesis.

Why Differencing is Necessary

Differencing is performed because:

1. **Trends:**
    - Differencing removes long-term linear trends (increasing or decreasing over time), making the data stationary.
2. **Seasonality:**
    - Repeating seasonal patterns are eliminated by applying differencing at specific lags (e.g., weekly or yearly cycles).
3. **Autocorrelation:**
    - Non-stationary data often exhibits high autocorrelation (i.e., strong relationships between lagged values). Differencing reduces autocorrelation, making the data more suitable for modelling.

6. PARTIAL AUTOCORRELATION AND AUTOCORRELATION FOR REGRESSION MODELS:

Why is Autocorrelation Important?

a. Identifying Patterns (Trends and Seasonality):
   - **Trends:** High autocorrelation at low lags indicates persistent values over time, signifying a trend.
   - **Seasonality:** Peaks in autocorrelation at specific lags indicate repeating cycles, such as daily, monthly, or yearly seasonality.

b. Model Selection and Parameter Estimation: Autocorrelation helps determine the structure of time-series models:
   - AR (AutoRegressive) Models: Use significant lags of the dependent variable as predictors. Autocorrelation indicates the relevant lags to include.
   - MA (Moving Average) Models: Depend on past error terms. Partial autocorrelation (PACF) helps identify the order.
   - ARIMA/SARIMA Models: Both autocorrelation (ACF) and PACF are used to identify the AR and MA terms and the need for differencing.

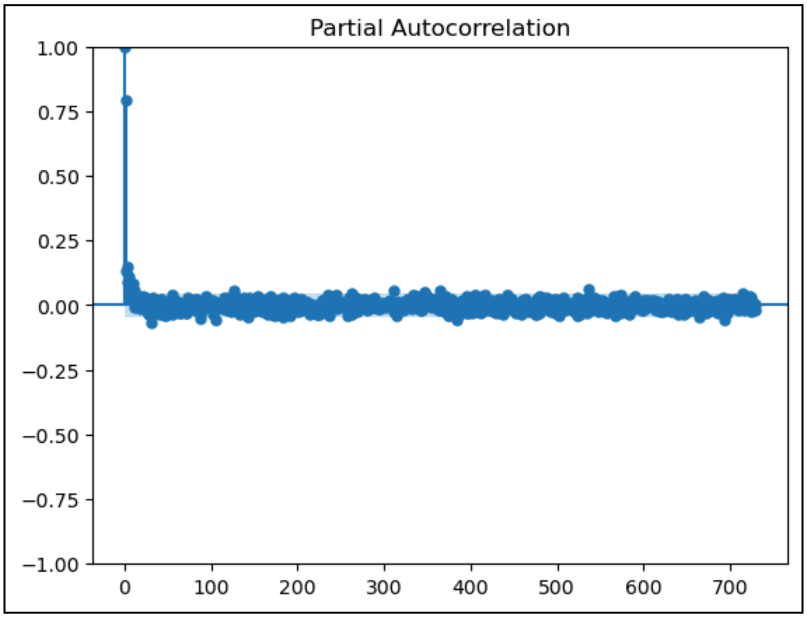
How is Autocorrelation Computed and Visualized?

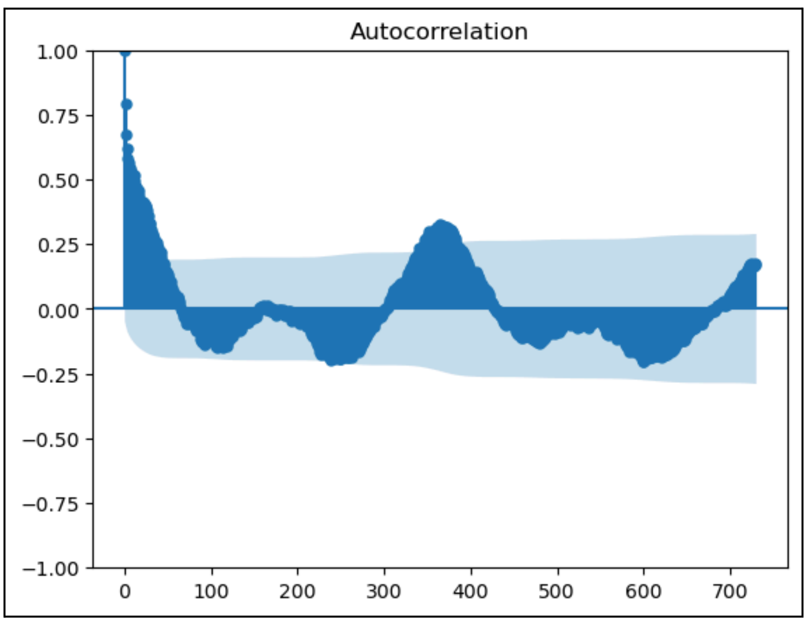
a. Autocorrelation Function (ACF):
- Plots the autocorrelation coefficients for different lags.
- Useful for identifying patterns like seasonality.

b. Partial Autocorrelation Function (PACF):
- Measures the correlation between Y_t and Y_{t-k}, removing the effects of intermediate lags(i.e., the lags between t and t−k).
- Helps identify the appropriate lag order for AR models.
- Daily Autocorrelation: The data is split into testing and training sets; and the data is tested for the last 365 days for both Jaipur and Kolkata.

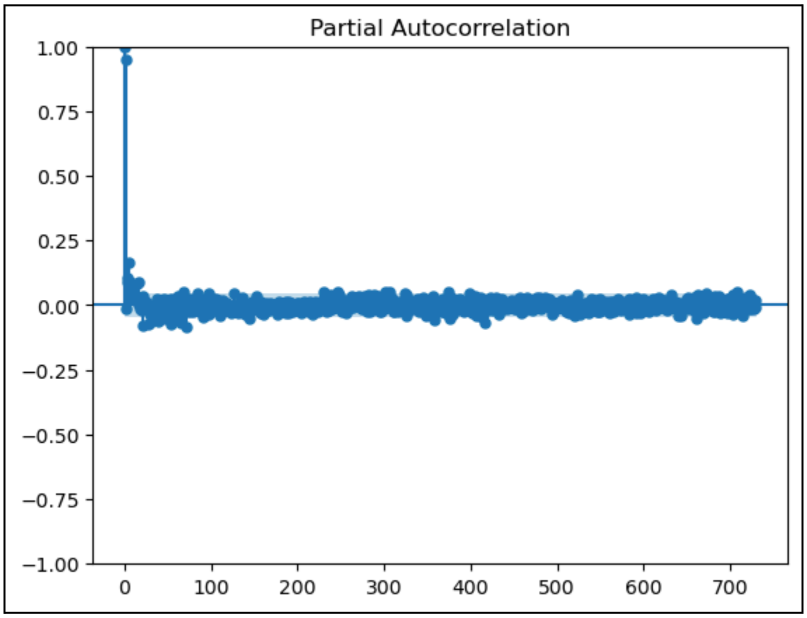
Daily Autocorrelation

1. **Jaipur:**

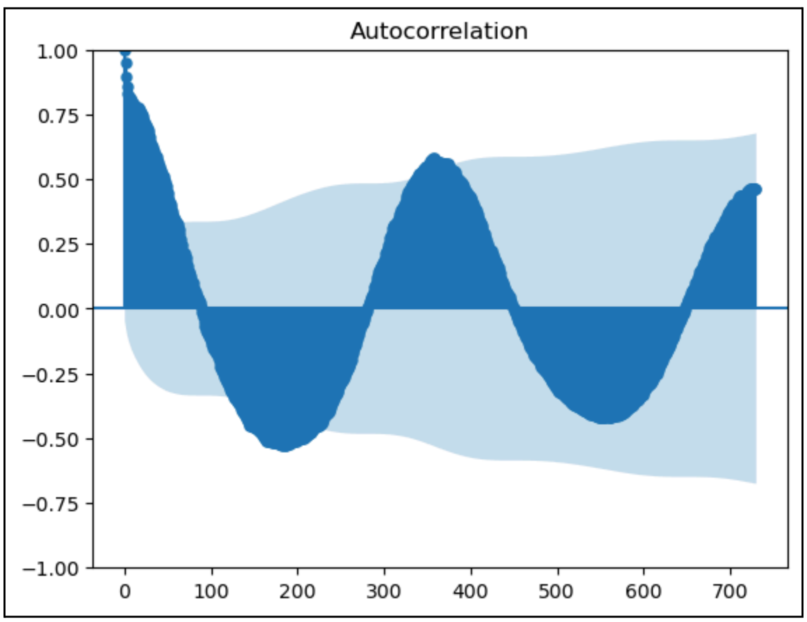




2. **Kolkata:**



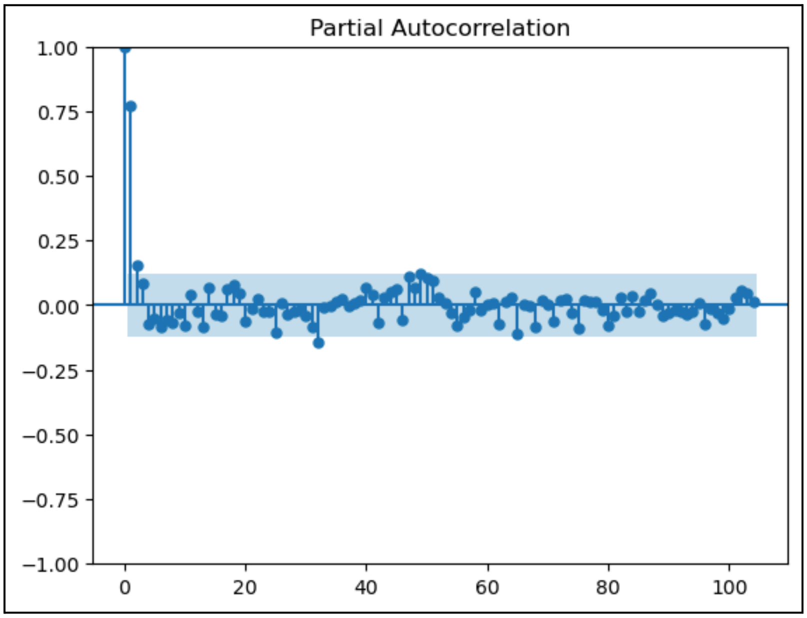
Partial Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
100
200
300
400
500
600
700

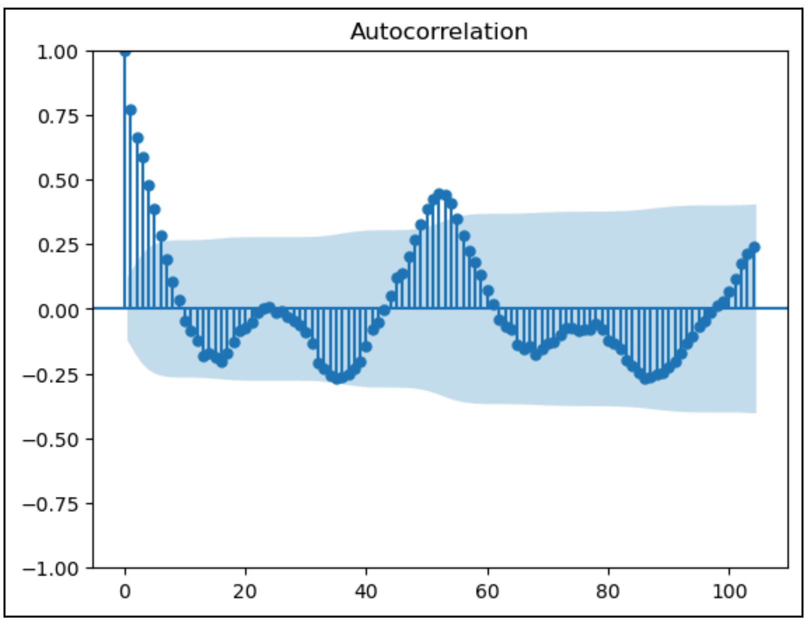


Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
100
200
300
400
500
600
700

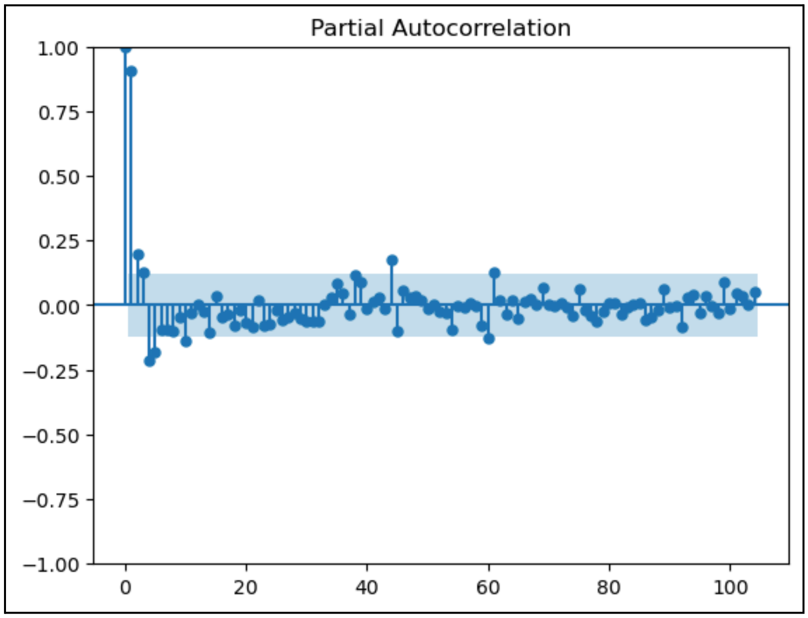
Weekly Autocorrelation:

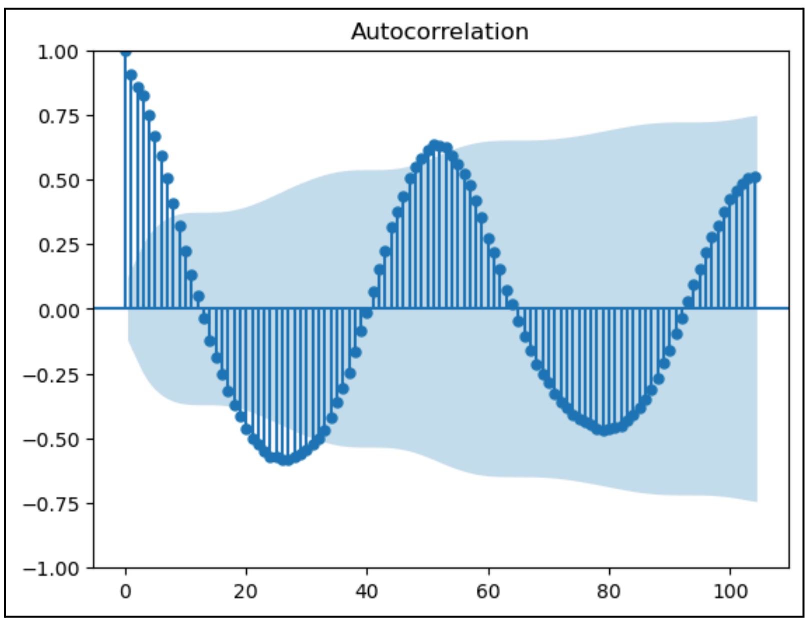
1. **Jaipur:**





2. **Kolkata:**





The ACF plots confirm that AQI data follows a yearly cycle, with the pattern being more prominent in Kolkata than in Jaipur.

7. TESTING AND VALIDATION OF REGRESSION MODELS

Models in Time-Series Analysis: The following regression models were used for the testing.

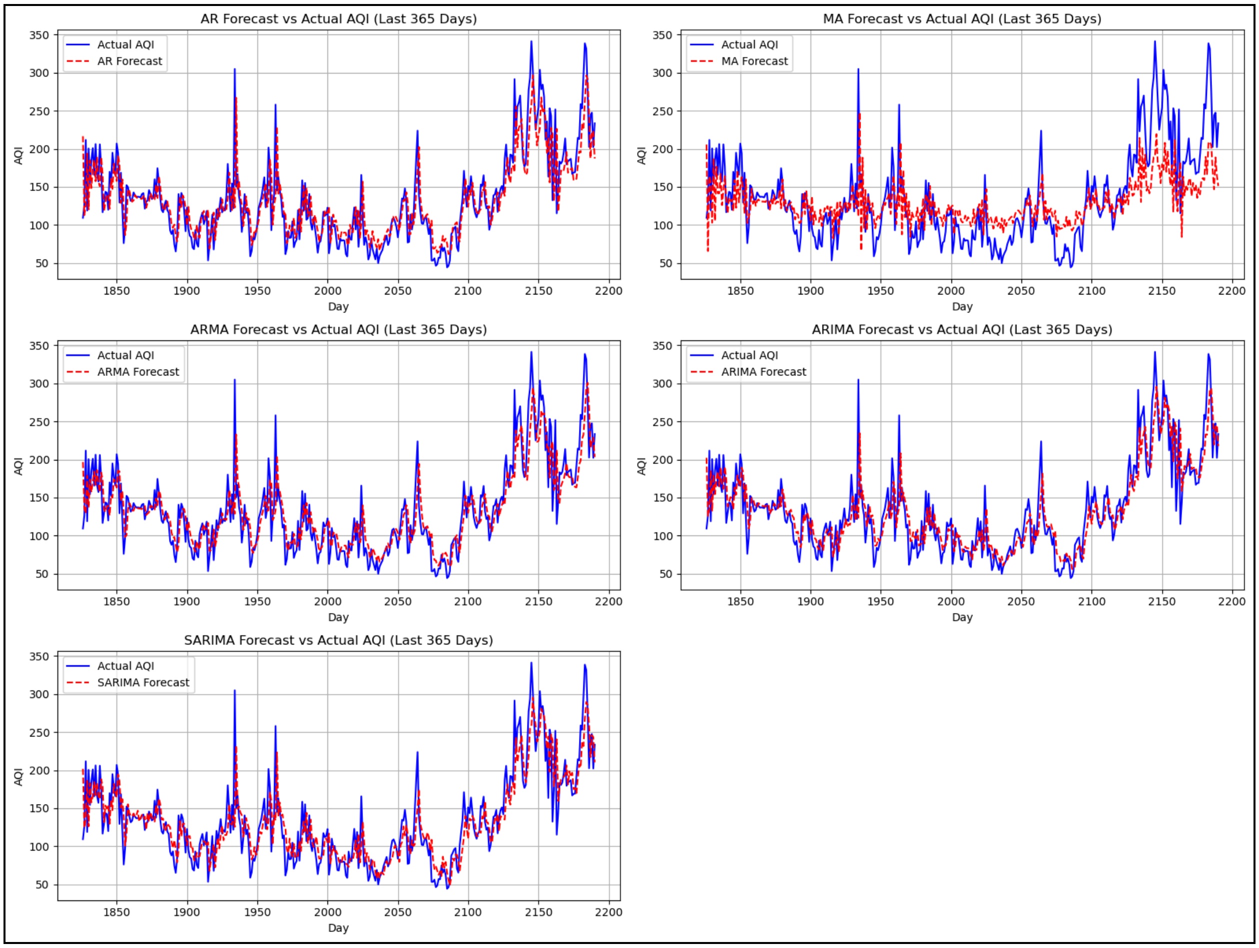
1. **AR (Autoregressive):** Uses past values of the time series to predict future values, relying on the relationship between lagged observations.
2. **MA (Moving Average):** Models the error terms (or noise) as a linear combination of past forecast errors.
3. **ARMA (Autoregressive Moving Average):** Combines AR and MA models to capture both lagged relationships and error patterns in a stationary time series.
4. **ARIMA (Autoregressive Integrated Moving Average):** Extends ARMA by including differencing to make non-stationary data stationary.
5. **SARIMA (Seasonal ARIMA):** Enhances ARIMA by including seasonal components (e.g., seasonal trends or cycles) for better performance on data with periodic patterns.

Validation Metrics: These validation metrics were used.

1. **R^2 (Coefficient of Determination):** Measures how well the model explains the variance in the data, with 1 indicating perfect prediction.
2. **RMSE (Root Mean Square Error):** Assesses the average magnitude of prediction errors, emphasizing larger errors due to squaring.
3. **MAE (Mean Absolute Error):** Computes the average absolute differences between predicted and actual values, offering a straightforward measure of accuracy.

Daily Forecasting and Validation

1. Jaipur



Model	R²	MAE	RMSE
AR Model	0.69	22.92	32.65
MA Model	0.44	32.15	43.71
ARMA Model	0.72	22.01	31.1
ARIMA Model	0.72	21.64	31.06
SARIMA Model	0.72	21.88	31.28

INFERENCE

Best Model:

- ARIMA demonstrates the best overall performance with the highest R^2 and the lowest MAE and RMSE, making it the most reliable model for forecasting.

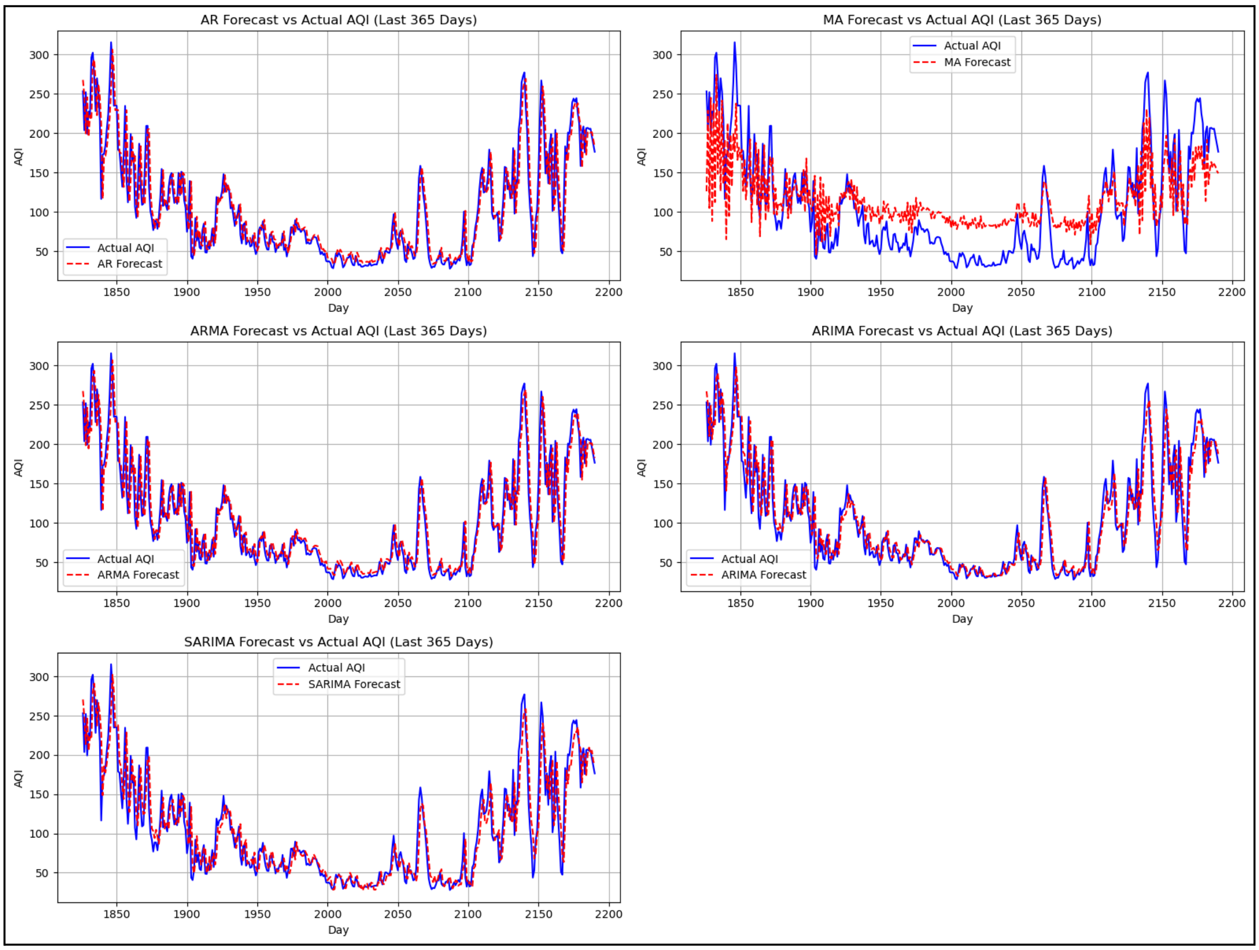
Seasonal Adjustment:

- SARIMA's performance is close to ARIMA, indicating that seasonal effects are not strongly dominant in the dataset.

Least Suitable Model:

- The MA model performs the worst, suggesting that modelling only the noise without accounting for lagged relationships is insufficient.

2. **Kolkata**



Model	R^2	MAE	RMSE
AR Model	0.85	18.12	26.18
MA Model	0.52	38.59	46.78
ARMA Model	0.85	18.1	26.17
ARIMA Model	0.85	17.75	25.9
SARIMA Model	0.85	18.21	25.95

INFERENCE

Best Model:

- ARIMA stands out as the most accurate and reliable model, with the lowest MAE (17.75) and RMSE (25.90).

Seasonality Handling:

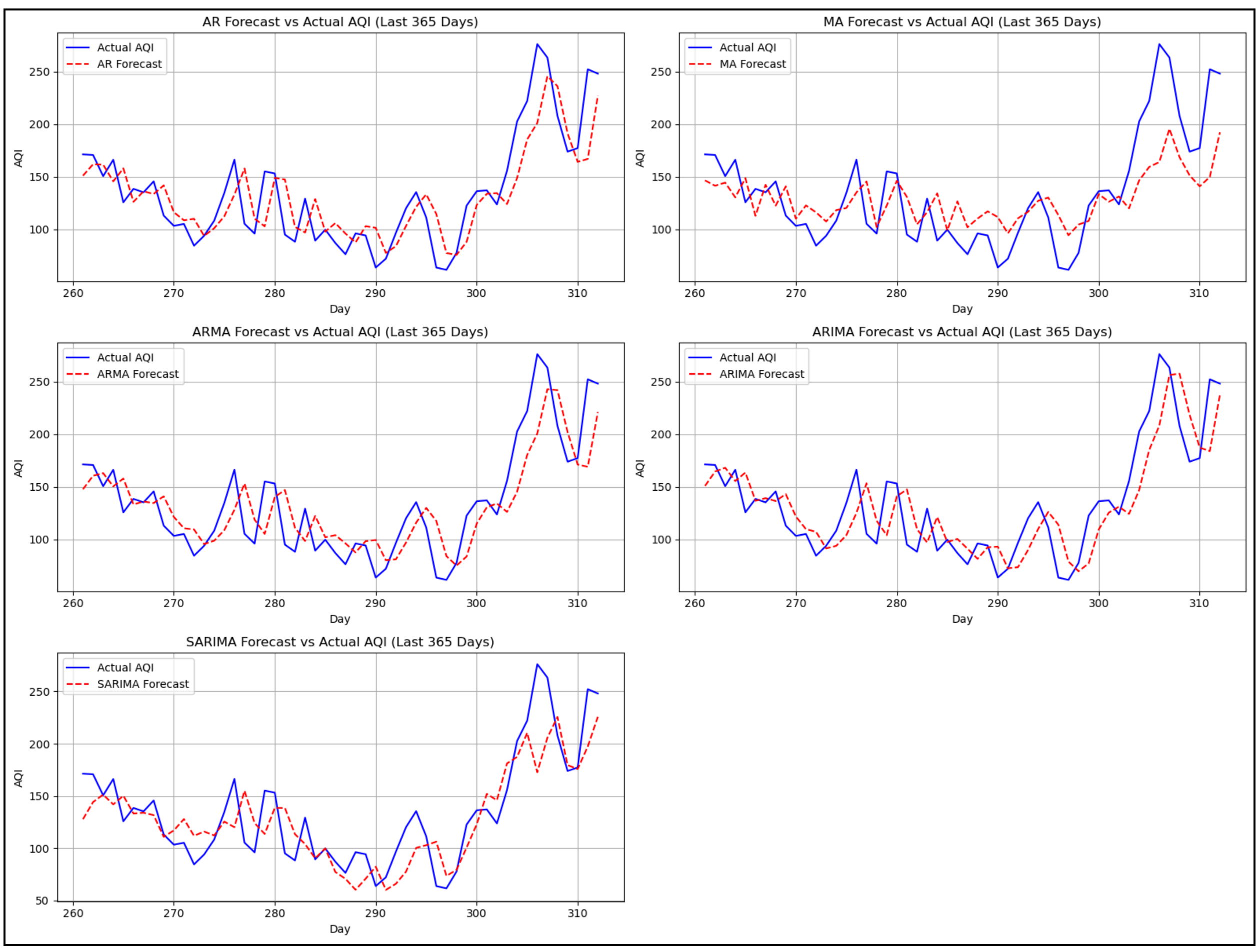
- SARIMA performs similarly to ARIMA, suggesting seasonal adjustments may not significantly improve forecasts for this dataset.

Least Suitable Model:

- The MA model performs poorly, with the highest MAE and RMSE, reaffirming that it is inadequate for capturing the underlying structure of the data.
- High Performers: AR and ARMA models also show strong performance, offering alternative choices with negligible differences in metrics compared to ARIMA.

Weekly Forecasting and Validation

1. Jaipur



Model	R^2	MAE	RMSE
AR Model	0.68	22.87	29.37
MA Model	0.52	28.45	36.23
ARMA Model	0.66	24.35	30.29
ARIMA Model	0.67	24.28	30.05
SARIMA Model	0.68	22.74	29.52

INFERENCE

Best Model:

- SARIMA emerges as the most reliable model, with the lowest MAE (22.74) and RMSE (29.52), effectively handling potential seasonal components.

Close Contenders:

- ARIMA and AR models perform well, with metrics close to SARIMA, suggesting their suitability for datasets with minimal seasonality.

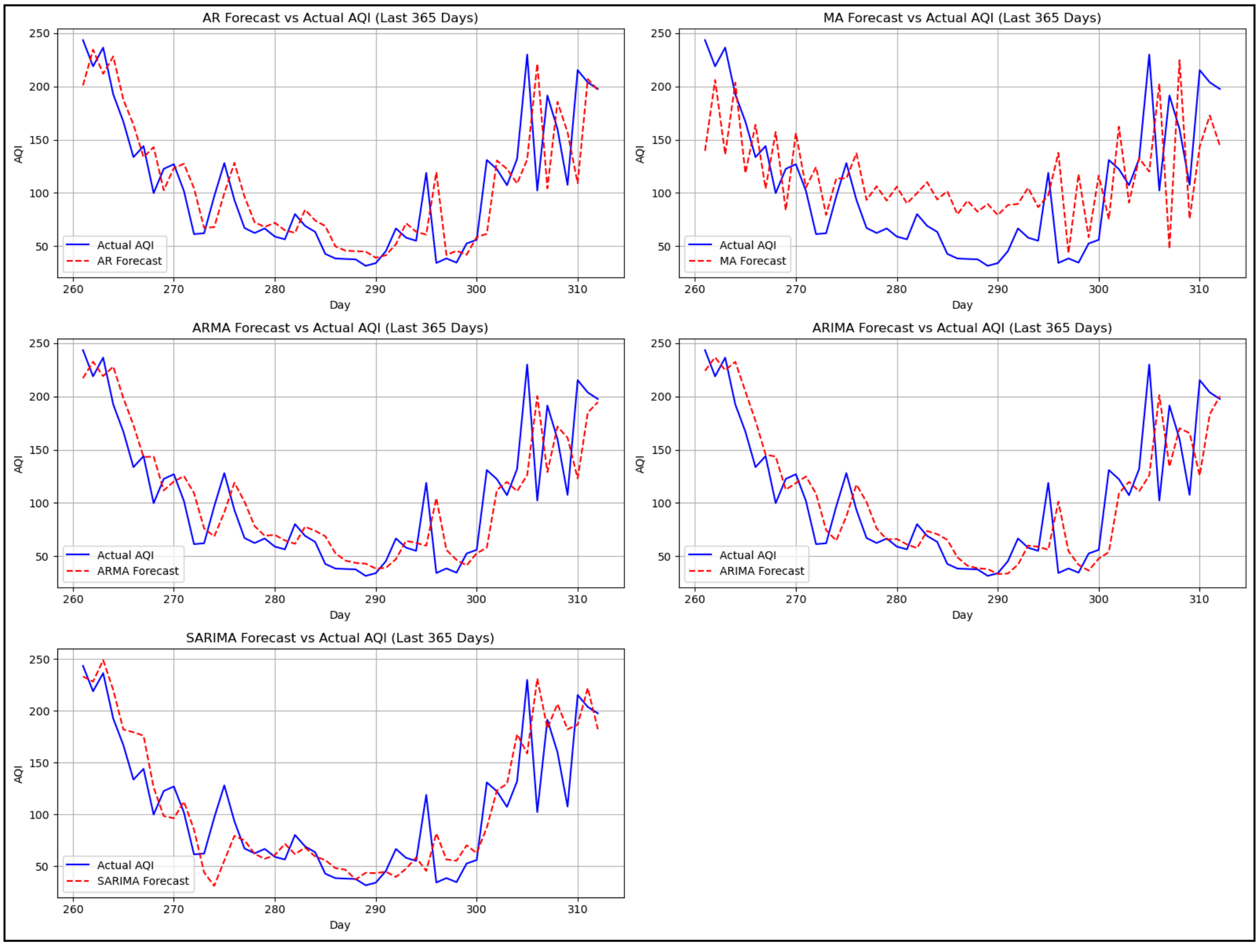
Least Suitable Model:

- The MA model has the weakest performance, with the lowest R^2 and highest error metrics, making it unsuitable for accurate forecasting.

ARMA Performance:

- While ARMA is effective, its slightly higher errors compared to SARIMA and ARIMA suggest it captures trends less effectively.

2. Kolkata:



Model	R²	MAE	RMSE
AR Model	0.6	27.05	38.93
MA Model	0.22	45.07	54.21
ARMA Model	0.66	26.04	36.13
ARIMA Model	0.65	25.71	36.43
SARIMA Model	0.68	24.61	35.01

INFERENCE

Best Model:

- SARIMA is the most effective model, achieving the highest R^2 (0.68) and the lowest MAE (24.61) and RMSE (35.01). It is well-suited for capturing seasonal patterns in the data.

Close Contenders:

- ARIMA and ARMA provide reasonably accurate predictions but are outperformed by SARIMA in all metrics.

Least Suitable Model:

- The MA model performs poorly, with the lowest R^2 and highest errors, making it unsuitable for forecasting this dataset.

AR Model Performance:

- While AR performs better than MA, it is less effective than SARIMA, ARIMA, and ARMA.

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